

Abhigna Kandala

Software Engineer | Backend, Cloud & AI Systems

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PROFESSIONAL SUMMARY

M.Eng. Software Engineering candidate (4.00 GPA) who builds and deploys cloud-native backend and full-stack systems with Python, TypeScript, React, Amazon Web Services (AWS), Docker, and Kubernetes. Co-developed a production-style stock-intelligence platform processing about 4,500 equities across five market universes; has built REST APIs, automated workflows, Continuous Integration/Continuous Deployment (CI/CD) pipelines, and cloud infrastructure. Prior SDE and data internships (Python, Node.js, TypeScript, MySQL).

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C++

Backend and APIs: FastAPI, Flask, Node.js, Express.js, REST API design, JSON Web Token (JWT) authentication, role-based access control
Cloud and DevOps: AWS (EC2, ECS Fargate, Lambda, S3, RDS, Step Functions, EventBridge, IAM), Docker, Kubernetes, GitHub Actions, CI/CD, Linux, Git

Databases: PostgreSQL, MySQL, MongoDB, schema design

AI and Data: Pandas, NumPy, Scikit-Learn, LightGBM, PyTorch, TensorFlow, Anthropic Claude API, retrieval-augmented generation (RAG)

Testing and Engineering: Pytest, Vitest, code review, technical documentation, Agile/Scrum, SDLC, object-oriented design

PROFESSIONAL EXPERIENCE

Data Analyst Intern | Firstsource Solutions Ltd. | Remote

May 2024 - Dec 2024

- Developed Python automation for recurring data-processing workflows used by a cross-functional group of analysts and machine learning (ML) engineers, reducing manual effort and improving the reliability of scheduled jobs.
- Reviewed code through GitHub pull requests with actionable feedback aligned with SDLC standards, and authored technical documentation that improved code consistency and handoffs.

Software Development Engineer Intern | Savart | Hyderabad, India

Jun 2023 - Nov 2023

- Built an internal research web interface from scratch with JavaScript, HTML, and CSS, including reusable UI components, form handling, and client-side state, so researchers could submit queries without external tools.
- Developed REST APIs with TypeScript and Node.js, integrated backend services with MySQL, and improved reliability of data retrieval; delivered features in Agile sprints with planning, standups, and code review on a distributed team.

Software Engineering Intern | Skolar | Remote

Oct 2022 - Dec 2022

- Implemented and documented Python data-structure and algorithm modules on Linux and integrated them into the project codebase; developed TypeScript code with unit tests and Docker containers, participating in code reviews under SDLC practices.

PROJECTS

T20 Predictor: Cloud-Native Stock Prediction Platform (Jan - Jun 2026) | Python, Flask, React, TypeScript, LightGBM, AWS

- Co-built, as one of two engineers reporting to the tech lead, a cloud-native platform that scores about 4,500 tickers across five market universes with LightGBM models and serves daily Top-20 stock predictions through a live web dashboard.
- Automated the end-to-end data and prediction workflow (download, dataset creation, feature engineering, inference, statistics) on AWS ECS Fargate, Step Functions, and EventBridge, replacing manual daily execution.
- Developed the Flask REST API over Amazon S3 datasets and RDS MySQL and built the React/TypeScript dashboard it powers; improved error handling and data fetching, expanded production coverage from three to five universes, and added Russell 1000 processing with 135,000+ feature rows across 451 tickers.
- Backtested a proposed swing-trading strategy and delivered a no-go recommendation (0.84 profit factor, 32.7% win rate), preventing further engineering investment in an unpromising direction.
- Supervised and code-reviewed four software interns, produced a prioritized 10-item review, and created a strict-TypeScript reference implementation with handoff documentation.

Terrapin Events: Campus Event Management System | React, TypeScript, FastAPI, MongoDB, Docker, Kubernetes, GitHub Actions

- Built, with a graduate software design team, a full-stack event platform with a FastAPI backend exposing 27 REST endpoints and a 14-page React/TypeScript frontend.
- Implemented JWT authentication and role-based access control for Admin, Organizer, and Participant workflows, including event approval, registration, automated waitlists, QR-code tickets, confirmations, and feedback requests.
- Wrote 110 Vitest and 94 Pytest tests (87% backend coverage); configured GitHub Actions CI/CD to build and push Docker images and deploy to Kubernetes with Nginx ingress, cert-manager TLS, secrets management, and autoscaling from 2 to 10 replicas.

Digital Time Capsule: Multimedia Sharing Web App | React, Node.js, Express.js, MongoDB, AWS EC2, Nginx

- Built a MERN-stack application for creating and sharing multimedia time capsules with friend requests, contributor permissions, privacy controls, authentication, and media storage.
- Implemented 25+ REST endpoints with JSON-schema validation across five MongoDB collections and an eight-page React frontend with reusable components and dynamic state; deployed to AWS EC2 behind Nginx and owned the full Linux-based release path.

EDUCATION

University of Maryland, College Park | M.Eng., Software Engineering, GPA 4.00 / 4.00

Jan 2025 - Dec 2026

Kalinga Institute of Industrial Technology (KIIT) | B.Tech, Computer Science and Engineering, GPA 8.81 / 10.00

2020 - 2024

CERTIFICATIONS AND ACHIEVEMENTS

AWS Training (Amazon Web Services) | HackerRank Problem Solving | Full Stack Web Development (Edu-Versity) | 250+ LeetCode problems